

The **User State Migration Tool** is one of the many tools provided in the Windows Assessment and Deployment Kit: <https://learn.microsoft.com/en-us/windows-hardware/get-started/adk-install>

A good use-case for this tool is when a user has a very busy schedule or is just tough in general to pin down for a PC swap. This tool allows us to copy user profiles without the need for them to sign in to the machine.

Script Location: I:\Scripts\User State Migration Tool-x64

Since this tool requires you to be an administrator and needs paths as arguments the easiest way I've found to run it is to copy the script folder to the computer you're working on locally instead of trying to use it from the network drive.

After clicking into the User State Migration Tool-x64 folder you'll see a couple of sub-items. The scripts folder contains some helper scripts that the usmt.exe wrapper tool utilizes, I'll go over usmt.exe at the end. The USMT/ folder contains everything the main tool needs to function properly. The MigCustom.xml file is the config file that the usmt.exe wrapper uses.

Navigate into the USMT/ folder with an administrator privileged terminal window to run the scanstate and loadstate programs. The tool is broken up into these two main programs, one for the computer to be migrated and one for the machine that is being deployed.

Starting with scanstate:

This is the part of the tool that captures the current state of the machine to be migrated, run this on the machine before it is swapped out.

There are a few arguments you need to pass into the command:

- **Store Location:** The directory path where the migration store will be created
- **XML Config File:** The file that defines the details of the migration / what to migrate
- **User Exclude Flag:** Use this flag to exclude certain users from the migration
- **User Include Flag:** Use this flag to include users in the migration explicitly
- **Follow Up Flags:** Flags like muting errors log file locations and verbosity can also be included

Here's an example of running scanstate.exe with each argument broken down:

```
.\scanstate.exe C:\Users\hgraves\Desktop\store /i:MigCustom.xml /ue:*\  
/ui:COLUSACOUNTY\* /c /o /v:13 /l:scan.log
```

The first argument is the migration store location, the .MIG file will be placed in the store folder (it's important to give the script a store folder, this comes into play with the loadstate script)

/i – The XML file that defines the details of the migration

/ue – The User Exclude flag, in the example shown I used wild cards to select all users such that I can specifically include what I want with the following flag. The format for users is DOMAIN\user

/ui – The User Include flag, in my example I am selecting all users under the COLUSACOUNTY domain with another wildcard. COLUSACOUNTY*

/c – Allows the script to continue execution regardless of any errors

/o – Overwrites data if applicable (log file locations / migration store locations)

/v – The verbosity level of the script

/l – The log file location

After this command finishes execution, you should have a store folder on your desktop with a .MIG file inside it, the loadstate.exe tool needs this whole folder as an argument, not just the .MIG file.

Move this store folder wherever you would like in order to get it to the new machine (flash drive, network drive temporarily, etc)

Once on the new machine, copy the User State Migration Tool to the computer again. Place the store folder from scanstate on the Desktop. Run the loadstate.exe tool with an admin terminal and the following arguments:

- **The Store Location:** The location of the store folder that contains the .MIG file
- **XML Config File:** The config file to pass in
- **Match User Flag:** This determines the users to be included from the migration file (in the context of a brand new computer, technically I don't think we even need this argument)
- **Follow-up Flags**

Here is an example of running the loadstate.exe command:

```
.\loadstate.exe C:\Users\hgraves\Desktop\store /i:MigCustom.xml  
/mu:COLUSACOUNTY*:COLUSACOUNTY* /c /v:13 /l:load.log
```

After running this command the new machine should be good to go and when the user signs in all of their data should be in place, if you use the MigCustom.xml file (or your own) their internet bookmarks/favorites for Edge are also included.

There is a lot I have not explored yet with this tool and others in this suite (Windows Assessment and Deployment Kit). There are many configurations and setups you can make with the use of the XML files allowing fine-tuned migrations if you want them.

More information and in-depth documentation can be found here:

<https://learn.microsoft.com/en-us/windows/deployment/usmt/usmt-overview>

If you want to be lazy:

I wrote a quick wrapper tool for these scripts to simplify the process for myself. After copying the User State Migration Tool-x64 folder to the target computer, run the **usmt.exe** program as administrator. A custom prompt will appear, and you'll have access to 2 main commands, **scan** and **load**. Type scan to run the scanstate.exe tool and grab all domain users, their data, and their bookmarks (it uses the MigCustom.xml file). The store will be placed on the desktop of the user running the script (you).

Move the store folder to the new machines desktop (this is the location where the load command is going to look by default) run usmt.exe as admin again and run the load command. This will execute loadstate.exe and use the MigCustom.xml file again to load the migration store to the target machine.